

Hussein Melhem

Phone:

+96171582819

Email:

Hussien.melhem81@gmail.com

Github:

<https://github.com/HusseinMelhem>

Portfolio:

<https://husseinmelhem.github.io/My-portfolio/>

EDUCATION

Lebanese University Faculty of Science | 2025-2026

Master 1 of Computer Science

- Relevant Coursework: Machine Learning, Networking, Cloud Computing, Image Processing, Distributed Systems.

Lebanese University Faculty of Science | 2021-2024

Bachelor of Computer Science

- Relevant Coursework: C, Data Structures, Java, Database, Calculus, Operating System, Parallel Programming.

PROFESSIONAL EXPERIENCE

Money Over Line | July 2024- Sep 2025

Software Engineer

- Designed and integrated an OCR-based ID data extraction system using agentic AI to enhance automation accuracy.
- Developed the complete backend architecture supporting both desktop and mobile applications.
- Contributed to the desktop application across both backend logic and application-level integration.

Skills: Python, FastAPI, Spring Boot, PostgreSQL, React, Delphi, Financial Systems, KYC/AML, Langchain

Smartec Technologies | Aug 2023- May 2024

Embedded Systems Software Engineer

- Coded in C FOC and SVPWM motor control algorithms, and controlled the functioning of the LCD.
- Created a desktop application that sends compressed images to the LCD of the motor.

Skills: STM32 Microcontrollers- Algorithms- Delphi desktop applications- I2C- C language- Pascal language

FNM Mall Freelance Project | Feb 2023- April 2023

- Implemented a full cart system with save for later functionality.
- Developed 2 types of checkouts either credit card or cash upon delivery.
- Created a full admin page for adding/editing items seeing orders

Skills: Laravel, Php, Mysql, Stripe

PERSONAL PROJECTS

Asset Allocation Performance Forecasting | Qube RT Data Challenge (Ongoing)

Built a next-day portfolio classification model using 79 engineered financial features and 32 LSTM embeddings, ensembled with CatBoost, XGBoost, and LightGBM, achieving 16th/831 (Top 2%) on the public leaderboard.

Hedge fund - Time Series forecasting | Kaggle Data Challenge (Ongoing)

Developed leakage-free sequential models trained per horizon, incorporating interaction features and subcategory-specific expert models, ranking 11th/1249 (Top 1%) on the public leaderboard.

Kaggle Playground Predicting Student Test Scores Data Challenge - Ranked 15/4282.

- Developed an Xgboost classifier to classify student grade into 1 of 7 quantile regions.
- Trained an Xgboost regressor on 58 other regressors(experts) then gated the output using the classifier's output.

Agentic AI Engineering Team

- Built a LangGraph AI engineering team where agents act like real PMs, developers, and QA engineers working together.
- Set up clear handoffs, reviews, and iteration loops so work moves naturally across the software lifecycle.

LLM Semantic Financial Analysis Platform

- Built a Web dashboard visualizing sector performance vs S&P 500 with next-day prediction and News API integration.
- Integrated Multi-LLM semantic analysis (Claude, GPT-4o, Ollama via OpenRouter) with chatbot for interactive insights.

ORGANIZATIONS

IEEE Student Branch | 2023-2024

Social Media Manager

- Participated in organization and display of events. Also gave a 12 sessions full stack web development boot-camp.

ACHIEVEMENTS

- Ranked 4th on Lebanon, 53 regionally, in IEEE xtreme 2022 global coding competition.

LANGUAGES

- English: C2
- French: A2
- Arabic: C2